



Stealth Press 1S — heat set insert press



Iconic FAB

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Summary

The most popular DIY heat set insert tool — now upgraded with the new 1S version

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About Stealth Press 1S

Stealth Press 1S: Latest version of precision DIY Heat Set Insert Tool

Stealth Press 1S is a highly accurate, mostly 3D-printed heat set insert tool compatible with generic, TS100, and Pinecil soldering irons.

New Stealth Press 1S Features

- Tool-free quick release mounts - swap between heat set insertion and regular soldering in seconds

- Interchangeable build plate inserts (open/closed designs) for better small part handling
- Built-in carrying handle for improved portability

Optional storage drawer keeps soldering tips and heat set inserts organized — [download here](#).



Core Benefits:

- Professional precision by 300mm MGN12H linear rail
- Integrated printable build plate
- Flexible retraction options - premium KEY-BAK or cost-effective alternatives
- Innovative design improvements over similar DIY solutions
- Professional results without expensive components

Bill of Materials and Sourcing guide

Buy kits

Vendor	Kit version	Product page	Region
KB3D.com	1S	Buy here	US

Veetec3D	1S	Buy here	EU, INT
DREMC	1S	Buy here	AU
Blurolls3D.com	1	Buy here	INT
Blurolls [on AliExpress]	1	Buy here	INT
Toaiot [on AliExpress]	1	Buy here	INT

R1 kits are **almost fully compatible** with R1S version, except 1 bolt and 1 nut, see Bill of Materials below for details.

Self sourcing guide

Part	Size / Standard	QTY	Source
M3 bolts	M3x8 SHCS	40	Buy here
M3 bolt [1]	M3x25 (or 30) SHCS	1	Buy here
M3 nut [2]	M3	1	Buy here
M5 bolts	M5x25 SCHS	2	Buy here
Bearings (F = flanged)	F695-2RS	4	Buy here
M3 threaded heat set inserts (M3*OD: 5*L:4)	M3x5x4	30	Buy here

Linear rail with bearing	MGN12H 300mm	1	Buy here
Retractable device (Key-Bak®)	Super48 36" 13oz	1	Buy here
Retractable device (Budget savvy alternative)	500g version		Buy here

Notes for R1S Bill of Materials

[1] [2] — 1x M3x25 (or 30) SHCS bolt and 1 x M3 nut is only parts difference between R1 and R1S versions.

Soldering irons and adapters

Soldering irons (supported variations)	Type	Source
Variation 1	Generic	Buy here
Variation 2	TS100	Buy here
Variation 3	Pinecil	Buy here
Heat insert adapters and tips	Type	Source
Soldering iron adapters and Tips (Generic)	Generic	Buy here
Soldering iron adapters and Tips (TS100 or Pinecil)	TS100 or Pinecil	Buy here

Alternative versions and add-ons

[Download printed parts for Budget savvy retractable device alternative](#)

[Drawer add-on](#)

Assembly manual

There is no specific Stealth Press 1S assembly manual at this time. However, assembly for Stealth Press 1S is fairly similar to version Stealth Press 1.

- [Download Stealth Press 1 assembly manual](#) with visuals and a [step-by-step guide](#), or;
- Follow instructions Iconic FAB's [Stealth Press 1S video build guide](#).

Video build guides

Iconic FAB video build guide (for 1S)

<https://www.youtube.com/watch?v=reMUZEUKHd0>

Other build guides (for version 1)

- Live build by Nero3D (for v1) — [Heatset Press and Fixing things](#)
- Live build by ModBotArmy (for v1) — [Building The Stealth Press Heat Insert Tool!](#)

Supported soldering irons

[Generic](#), [TS100](#) or [Pinecil](#).

Print guidelines

Here are print guidelines for you to follow in order to have decent success with printing parts:

3d printing process	FDM (fused deposition modeling)
Infill type	Support cubic, Grid, Gyroid, Honeycomb, Triangle or Cubic

Material	ASA, ASA-CF, ABS, ABS+, ABS-CF [1]
Infill percentage	25 - 60% [2]
Layer height	0.2mm, recommended
Wall count	Recommended: 4
Solid top/bottom layers	Recommended: 5
Extrusion width	0.4mm, recommended

Side notes to the print guidelines

[1] PLA, PLA+, and PETG are also considerable filament choices. However, take into account that soldering iron radiates some indirect heat. **The usage of less temperature-stable filaments may result in part deformations over time.** It is recommended to use PETG if ideally - ASA or ABS at least for “arm” and “iron mount” parts.

[2] Consider using **60% or more** of infill for **build plate** part to ensure it's not sagging over time and 25-40% for all other parts.

A wall count of 4 is an absolute must to ensure the structural integrity of the parts.

File naming for printed parts

The parts use this naming convention:

Primary color	These files will have nothing at the start of the filename. Example: buildplate_x1.stl
Accent color	Files have “[a]” to the front of any STL file that is intended to be printed with accent color. Example: [a] _bushing_x1.stl

Count of parts	Files have “ x1 ” to “ x4 ” at the end of any STL file. The number represents the count of identical parts to be printed . Example: [a] _m5_1mm_spacer_x4 (In this case you will need to print 4 identical spacers).
Revision number	Files have i.e. “ _r1s ” suffix at the end of any STL file. This number is file revision number.

How to support the project

Here are most popular choices how to do so:

- **Consider becoming member of Iconic FAB Club** on Printables and **get all Paid models and Manuals for single membership fee**, starting from as low as \$3.
- **Press like** and **download** model files. It helps more than I can express.

Special thanks

- To **@joshmurrah** for ginormous assistance in testing and improvements for 1S version and **generally being amazing human being**
- To **@radicalreel** for his version 1 mod and inspiration for carrying handle
- To **@3DPMAMSIH** for his version 1 drawer mod and ideas which lead to creating proprietary drawer add-on for 1S version.
- To **all great 3DP community** building project, posting comments and makes, following me in social media **and supporting what I do in so many amazing ways**.

Credits and acknowledgements

Here is the list of other open-source projects used for sourcing multiple CAD parts.

Listed models below are modified to fit the project purpose and design:

- VORON Switchwire: <https://github.com/VoronDesign/Voron-Switchwire> [Base for remix of KEY-BAK mounting parts]
- Ender SW: https://github.com/boubounokefalos/Ender_SW [Base for remix of Support arms]

Change log

Date	Changes
03.12.2024	soldering_iron_mount_generic_ID18_x1_r1s.stl — updated to correct ID size of 18mm
26.11.2024	R1S release

Model files



top_handle_x1_r1s.stl



center_x1_r1s.stl



leg_x1_r1s.stl



buildplate_x1_r1s.stl



a_buildplate_bushing_closed_x1_r1s.stl



a_buildplate_bushing_open_x1_r1s.stl



a_strain_relief_r_x1_r1s.stl



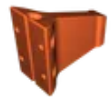
a_strain_relief_l_x1_r1s.stl



support_arm_l_x1_r1s.stl



support_arm_r_x1_r1s.stl



arm_x1_r1s.stl



keybak_plate_a_x1_r1s.stl



keybak_plate_b_x1_r1s.stl



a_m5_1mm_spacer_x4_r1s.stl



a_soldering_iron_mount_knob_x1_r1s.stl



soldering_iron_mount_ts100_x1_r1s.stl



soldering_iron_mount_pine_x1_r1s.stl



soldering_iron_mount_generic_id20_x1_r1s.stl



soldering_iron_mount_generic_id19_x1_r1s.stl



soldering_iron_mount_generic_id18_x1_r1s.stl



soldering_iron_mount_generic_id17_x1_r1s.stl



soldering_iron_mount_generic_id16_x1_r1s.stl



soldering_iron_mount_generic_id15_x1_r1s.stl



soldering_iron_mount_generic_id20_x1_r1s.step

☐ STEP file for generic soldering iron mount (20mm ID)

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